

RESEARCH PROPOSAL

GRPO-Based Reinforcement Learning for Flow-Matching VLA Models

*Applying Group Relative Policy Optimization to SmolVLA
for Improved Generalization in Robotic Manipulation*

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1. Introduction

Recent progress in Vision-Language-Action (VLA) models has demonstrated strong capabilities in robotic manipulation tasks, primarily driven by supervised imitation learning on large-scale demonstration datasets. More recently, SimpleVLA-RL [3] has shown that finetuning VLA models with Group Relative Policy Optimization (GRPO) yields promising improvements beyond pure imitation. However, these approaches still rely on autoregressive generation over discrete action tokens, which limits both the precision of the resulting control signals and inference efficiency — a particularly restrictive choice for continuous, high-frequency manipulation.

This proposal addresses that gap by introducing a reinforcement learning fine-tuning stage on top of a pre-trained SmolVLA [8] model, using Group Relative Policy Optimization (GRPO) as the training algorithm. SmolVLA [8] is a compact, open-source VLA architecture that employs a flow-matching action head rather than the autoregressive discrete token head used in many contemporary VLA systems. This architectural distinction makes applying RL non-trivial and represents the central technical contribution of this work.

The core hypothesis is that by generating multiple trajectory rollouts from the flow-matching head via **a branching mechanism that relies on a single initial noise without subsequent resampling (thereby reducing computational complexity)**, and computing outcome-based binary rewards, it is possible to construct a group-relative reward signal. This signal drives the policy toward higher task success rates while preserving the expressiveness of the learned action distribution.

Problem Statement

Supervised fine-tuning on robotic manipulation datasets produces policies that fail to generalize beyond the training distribution. This project investigates whether GRPO-based RL fine-tuning on SmolVLA’s flow-matching head can improve generalization without the instabilities typically associated with policy gradient methods on continuous action spaces.

2. Background and Related Work

2.1 The Generalization Problem in Imitation-Learned VLA Models

Imitation learning (IL) from expert demonstrations has been the dominant training paradigm for robotic manipulation. Models such as RT-2 [9], OpenVLA [2], Pi-0 [1] and SmolVLA [8] are pre-trained on web-scale vision-language data and then fine-tuned on robot trajectories via behavioral cloning. While this produces compelling in-distribution performance, the resulting policies are brittle they lack the ability to adapt to novel object configurations, or transfer to unseen environments.

The SimpleVLA-RL [3] paper formalizes this observation and proposes reinforcement learning as the corrective mechanism. By replacing the imitation signal with an environment-based reward, the policy is forced to discover generalizable strategies rather than replay memorized solutions. However, the paper’s experiments are primarily conducted on autoregressive token-based action heads, leaving flow-matching architectures unexplored.

2.2 Group Relative Policy Optimization (GRPO)

GRPO is a policy gradient variant originally proposed in the context of large language model reasoning DeepSeekMath [7]. Unlike PPO, which requires a separately trained value function, GRPO estimates the advantage of each sample relative to the group average from a single batch of rollouts. For a group of n outputs sampled from the same prompt, the advantage of output i is computed according to Eq. (1).

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$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \dots, r_n)}{\text{std}(r_1, \dots, r_n) + \epsilon} \quad (1)$$

This formulation is particularly attractive for robotics because it eliminates the need to train a critic network, which is notoriously difficult to stabilize over high-dimensional continuous observation spaces. The group-relative normalization also provides a natural curriculum when all trajectories succeed or all fail, the advantage signal is zero, and no update is made, which prevents reward hacking.

2.3 SmolVLA and the Flow-Matching Action Head

SmolVLA [8] is a compact VLA model built on the SmolVLM [6] vision-language backbone. Unlike VLA models that decode actions as discrete language tokens, SmolVLA [8] employs a flow-matching head (Also known as action expert) that models the action distribution as a continuous probability path. At inference time, actions are generated by integrating a learned vector field from a randomly sampled Gaussian noise vector over a fixed number of denoising steps (typically 10).

2.4 Flow Matching for Continuous Action Generation

Flow matching [4] is a generative modeling framework that learns a continuous-time vector field to transport samples from a simple prior distribution (typically a standard Gaussian) to a target data distribution. In contrast to diffusion models, which are trained against a stochastic denoising objective and sampled via a stochastic differential equation, flow matching directly regresses a deterministic velocity field along a prescribed probability path, yielding an ordinary differential equation at inference time.

Formally, given a target distribution p_1 over actions and a prior $p_0 = \mathcal{N}(0, I)$, flow matching defines a time-dependent vector field $v_\theta(x, t)$ parameterized by a neural network and trained to match the conditional velocity along a linear interpolation path between noise and data. This linear path, also known as the rectified or conditional optimal transport path, produces straight trajectories whose constant conditional velocity is efficient to learn and integrate. The training objective for a sampled action $a \sim p_1$ and noise $z \sim p_0$ is given by Eq. (2).

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1], a \sim p_1, z \sim p_0} \left[\|v_\theta(x_t, t) - (a - z)\|^2 \right] \quad (2)$$

where $x_t = (1 - t)z + ta$ is the linear interpolant between the noise vector and the target action, and the regression target $(a - z)$ is the corresponding conditional velocity dx_t/dt . At inference time, an action is generated by integrating the learned vector field forward in time from $t = 0$ to $t = 1$, starting from an initial noise sample $z \sim p_0$. In practice, this integration is performed using a fixed number of Euler steps N , producing the iterative update in Eq. (3).

$$x_{t+\Delta t} = x_t + \Delta t \cdot v_\theta(x_t, t), \quad \Delta t = \frac{1}{N} \quad (3)$$

The final action is recovered as $a = x_1$. This formulation has two properties that are particularly relevant for the present work. First, for any fixed integration budget N , the mapping from the initial noise z to the generated action is fully deterministic, so all stochasticity in the output is controlled by the choice of z . Second, the number of integration steps N is a free parameter at inference time and can be varied without retraining, enabling trade-offs between computational cost and action fidelity.

3. Research Objectives

This project pursues the following primary objectives:

1. Develop a principled GRPO training pipeline for flow-matching VLA architectures, using independent noise sampling as the source of policy stochasticity.
2. Design a weighted denoising reward scheme that provides denser reward signal by weighting intermediate denoising steps, reducing reward sparsity without requiring dense environment feedback.
3. Evaluate whether RL fine-tuning on SmolVLA [8] improves generalization and success rate to out-of-distribution manipulation scenarios compared to the IL baseline in libero [5] benchmark.

4. Methodology

4.1 Overview of the Training Pipeline

The proposed training pipeline follows the standard GRPO loop, Sample a group of n trajectories for each observation context, evaluate each trajectory against the task reward, compute group-relative advantages, and update the policy via a clipped policy gradient objective. The key adaptation for flow-matching is in how trajectories are sampled.

4.2 Trajectory Sampling via Independent Noise

Because SmolVLA’s flow-matching head is deterministic given a fixed noise vector, diversity across the n trajectories in a group must be induced by sampling n distinct Gaussian noise vectors at the start of each rollout. Formally, for a given observation o and language instruction l , the i -th trajectory is generated according to Eq. (4).

$$\tau_i = \text{Execute}(\text{FlowDenoise}(\pi_\theta(o, l), z_i)), \quad z_i \sim \mathcal{N}(0, I) \quad (4)$$

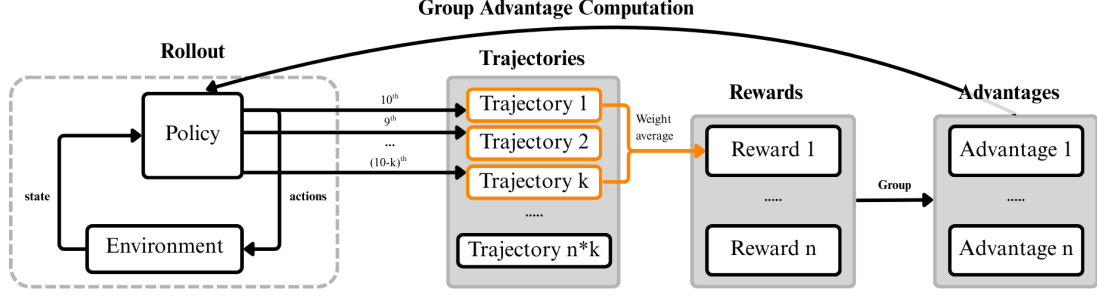


Figure 1: Overview of the proposed GRPO training pipeline for SmolVLA [8]. For each observation, the policy is rolled out with k independent noise vectors across multiple denoising horizons (8, 9, and 10 steps), producing $n \times k$ trajectories in total. Rewards from the three horizons are combined via a weighted average to form a single per-trajectory reward, and group-relative advantages are then computed across the n trajectories to update the policy.

Each noise vector z_i produces a distinct action sequence, which is then executed in the simulation environment to produce a full trajectory. The binary task success reward $r_i \in \{0, 1\}$ is assigned based on whether the trajectory achieves the specified manipulation goal.

Key Design Principle

Independent noise sampling plays the same role as stochastic token sampling in autoregressive VLA models. Without it, all n trajectories would be identical and the GRPO advantage signal would collapse to zero regardless of reward.

4.3 Weighted Denoising Reward

Binary trajectory rewards are highly sparse, the agent receives no feedback until a full rollout completes, and a single policy error at any step can result in zero reward for the entire trajectory. To mitigate this, the proposed method introduces a weighted denoising reward that aggregates task outcomes across multiple denoising horizons within the same execution to Eq. (5).

$$R(\tau_i) = 0.1 \cdot r(\tau_i^{(8)}) + 0.2 \cdot r(\tau_i^{(9)}) + 0.7 \cdot r(\tau_i^{(10)}) \quad (5)$$

Here, $\tau_i^{(n)}$ denotes the trajectory executed using n denoising steps from the same initial noise z_i and observation context. The intuition is that shorter denoising horizons (8-step, 9-step) produce noisier but structurally related actions if the policy is learning a genuinely effective strategy, it should succeed even at reduced denoising fidelity, whereas a policy that is exploiting a memorized trajectory will fail at reduced steps.

The weighting scheme assigns the majority of the reward signal (70%) to the fully-denoised 10-step trajectory, preserving the primacy of high-fidelity action execution, while the 8-step and 9-step components provide supplementary gradient signal that discriminates between robust and brittle strategies.

Table 1: Weighted denoising reward components used during training. Lower-step rollouts provide auxiliary gradient signal, while the full-horizon rollout dominates the total reward.

Denoising Steps	Weight	Interpretation
8 steps	0.1	Low-fidelity rollout; verifies structural correctness of the policy.
9 steps	0.2	Intermediate fidelity; supplies a mid-range gradient signal.
10 steps (full)	0.7	Full denoising horizon; primary reward signal for task completion.

4.4 GRPO Objective

Given a group of n trajectories with weighted rewards $R_1, \dots, R_n$, the group-relative advantage is computed according to Eq. (1).

The policy gradient update follows the clipped GRPO objective. Because the log-probability of a trajectory under the flow-matching head is defined with respect to the noise vector and the flow-matching loss over the denoising path, the gradient computation requires backpropagating through the denoising integration. In practice, this is implemented using the flow-matching score function estimator.

4.5 Fine-Tuning Strategy

Fine-tuning begins from the SmolVLA [8] checkpoint that has been pre-trained and supervised fine-tuned (SFT) on the libero [5] simulation. The RL stage is applied to the flow-matching head and the vision-language backbone jointly, with a lower learning rate on the backbone to preserve the pre-trained visual and linguistic representations. The SFT policy is also retained as a reference for KL-divergence regularization, preventing the policy from drifting too far from the imitation-learned prior, we optimize the objective defined as in Eq. (6).

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}\left[\min\left(\rho_i(\theta) \hat{A}_i, \text{clip}(\rho_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i\right)\right] \quad (6)$$

where $\rho_i(\theta) = \pi_\theta(\tau_i)/\pi_{\theta_{\text{old}}}(\tau_i)$ denotes the importance sampling ratio between the current policy and the policy used to collect the rollouts, and ϵ is the clipping threshold.

5. Scope and Limitations

This research is strictly bounded by the following parameters to ensure a focused and computationally feasible investigation:

- **Model Architecture:** The investigation is exclusively limited to the SmolVLA architecture. Other Vision-Language-Action (VLA) models or traditional autoregressive token-based action heads are not included in this comparative study.

- **Dataset and Training Environment:** All training phases, including both the supervised fine-tuning (SFT) baseline and the subsequent GRPO-based reinforcement learning, will be conducted solely on the LIBERO simulation datasets. Real-world robotic deployment and physical hardware testing fall outside the scope of this proposal.
- **Evaluation Benchmark:** The policy’s performance—encompassing task success rates, robustness, and out-of-distribution generalization—will be evaluated strictly using the standardized LIBERO benchmark suite.

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